

UR5: Manipulability and Inverse Kinematics

RBE 501

i	a_i [m]	α_i	d_i [m]	θ_i
1	0	$\pi/2$	$d_1 = 0.08916$	θ_1
2	$a_2 = 0.425$	0	0	θ_2
3	$a_3 = 0.39225$	0	0	θ_3
4	0	$\pi/2$	$d_4 = 0.10915$	θ_4
5	0	$-\pi/2$	$d_5 = 0.09456$	θ_5
6	0	0	$d_6 = 0.0823$	θ_6

1 Manipulability

$$w(\mathbf{q}) = \sqrt{\det(J(\mathbf{q}) J(\mathbf{q})^T)} = a_2 a_3 |s_3| |s_5| |a_2 c_2 + a_3 c_{23} + d_5 s_{234}| \quad (1)$$

Numerically, $a_2 a_3 = 0.425 \times 0.39225 = 0.16671 \text{ m}^2$, so

$$w(\mathbf{q}) = 0.16671 |s_3| |s_5| |0.425 c_2 + 0.39225 c_{23} + 0.09456 s_{234}|. \quad (2)$$

Singular configurations ($w = 0$)

1. **Elbow singularity:** $s_3 = 0 \Rightarrow \theta_3 = 0$ or π (the arm is fully stretched out or folded back).
2. **Wrist singularity:** $s_5 = 0 \Rightarrow \theta_5 = 0$ or π (the axes of joints 4 and 6 are parallel, so they lose a DoF).
3. **Shoulder singularity:** $a_2 c_2 + a_3 c_{23} + d_5 s_{234} = 0$ (the wrist center lies on the plane containing the axes of joints 1 and 2).

2 Inverse Kinematics

$${}^0T_6 = \begin{bmatrix} x_x & y_x & z_x & p_x \\ x_y & y_y & z_y & p_y \\ x_z & y_z & z_z & p_z \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad (3)$$

Joint 1 (shoulder left/right).

$${}^0\mathbf{p}_5 = {}^0\mathbf{p}_6 - d_6 {}^0\hat{\mathbf{z}}_6 = \begin{bmatrix} p_x - d_6 z_x \\ p_y - d_6 z_y \\ p_z - d_6 z_z \end{bmatrix}, \quad (4)$$

$$\theta_1 = \text{atan2}({}^0p_{5y}, {}^0p_{5x}) \pm \cos^{-1}\left(\frac{d_4}{\sqrt{{}^0p_{5x}^2 + {}^0p_{5y}^2}}\right) + \frac{\pi}{2} \quad (5)$$

This requires $\sqrt{{}^0p_{5x}^2 + {}^0p_{5y}^2} \geq d_4$.

Joint 5 (wrist up/down).

$$\theta_5 = \pm \cos^{-1}\left(\frac{p_x s_1 - p_y c_1 - d_4}{d_6}\right) \quad (6)$$

Joint 6. For $s_5 \neq 0$:

$$\theta_6 = \text{atan2}\left(\frac{-y_x s_1 + y_y c_1}{s_5}, \frac{x_x s_1 - x_y c_1}{s_5}\right) \quad (7)$$

If $s_5 = 0$, joints 2, 3, 4 and 6 are parallel. The solution is then underdetermined, so θ_6 may be chosen freely.

Joints 2, 3, 4 (planar 3R arm, elbow up/down).

$${}^1T_4 = ({}^0T_1)^{-1} {}^0T_6 ({}^5T_6)^{-1} ({}^4T_5)^{-1}, \quad (8)$$

$$\theta_3 = \pm \cos^{-1}\left(\frac{x_{14}^2 + y_{14}^2 - a_2^2 - a_3^2}{2a_2a_3}\right) \quad (9)$$

$$\theta_2 = \text{atan2}(y_{14}, x_{14}) - \text{atan2}(a_3 s_3, a_2 + a_3 c_3) \quad (10)$$

$$\theta_4 = \text{atan2}({}^1R_{4,21}, {}^1R_{4,11}) - \theta_2 - \theta_3 \quad (11)$$

No solution exists if $\sqrt{x_{14}^2 + y_{14}^2} > a_2 + a_3$ or $\sqrt{x_{14}^2 + y_{14}^2} < |a_2 - a_3|$.

Solution branches

Joint	Sign choice	Physical meaning
θ_1	$\pm \cos^{-1}(\cdot)$	shoulder left / right
θ_5	$\pm \cos^{-1}(\cdot)$	wrist up / down
θ_3	$\pm \cos^{-1}(\cdot)$	elbow up / down